

Engineering Mathematics III (ENGE702)

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Lecture 19: Fourier Series Applications with Exercises

Fourier Series

We will now revisit the mass - spring - damper systems, and see how Fourier series can help us analyse forced oscillations.

Our previous analysis consisted of using sinusoidal external forcing functions ($\sin$ or $\cos$), but in general the external force can be any function.

In particular, if it is a periodic function, we can represent it as a Fourier series.

Fourier Series

Example: Suppose we have a mass - spring - damper system with mass $m = 1$ kg, damping constant $c = 0.05$ kg/s, and spring constant $k = 25$ N/m. Suppose then the system has a periodic external force applied that is given by:

$$r(t) = \begin{cases} \frac{\pi}{2} + t & \text{if } -\pi < t \leq 0 \\ \frac{\pi}{2} - t & \text{if } 0 < t \leq \pi \end{cases}$$

and $r(t + 2\pi) = r(t)$. Find the steady state solution $y(t)$ (in metres).

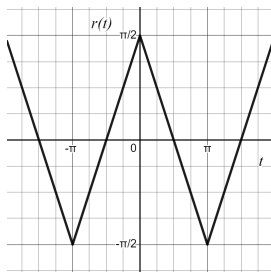


Figure: The periodic forcing function $r(t)$ (measured in newtons)

Fourier Series

Find the steady state solution $y(t)$ given

$$y''(t) + 0.05y'(t) + 25y(t) = r(t)$$

1. Consider the damping for the unforced system. Since mass $m = 1$, damping constant $c = 0.05$, and spring constant $k = 25$,

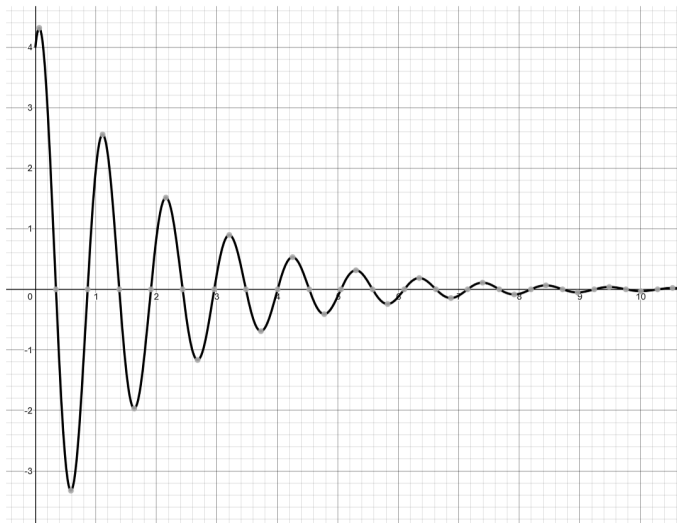
$$c^2 - 4mk = (0.05)^2 - 4(1)(25) < 0$$

and so we have damped oscillations that tend to zero as time $t \rightarrow \infty$. Hence the homogeneous solution $y_h(t) \rightarrow 0$ as $t \rightarrow \infty$ and we need only find the particular solution.

2. We need to find the sine and cosine terms in the Fourier series for $r(t)$, and for each $n \geq 0$, find a particular solution y_n .
3. The steady state solution $y(t)$ will be the sum of the y_n .

Underdamping $c^2 - 4mk < 0$

The solution to the unforced system will exhibit a shape that is something like this:



Fourier Series Calculations

Note that since $r(t)$ is an even function, all the $b_n = 0$. Here $L = \pi$. Find a_0 .

$$\begin{aligned}a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} r(t) dt \\&= \frac{1}{2\pi} \int_{-\pi}^0 \left(t + \frac{\pi}{2}\right) dt + \frac{1}{2\pi} \int_0^{\pi} \left(-t + \frac{\pi}{2}\right) dt \\&= \frac{1}{4\pi} \left(t + \frac{\pi}{2}\right)^2 \Big|_{-\pi}^0 + \frac{-1}{4\pi} \left(-t + \frac{\pi}{2}\right)^2 \Big|_0^{\pi} \\&= \frac{1}{4\pi} \left(\frac{\pi^2}{4} - \frac{\pi^2}{4}\right) - \frac{1}{4\pi} \left(\frac{\pi^2}{4} - \frac{\pi^2}{4}\right) \\&= 0\end{aligned}$$

Since the average height of the function is zero we would expect $a_0 = 0$.

Fourier Series Calculations

Find a_n , $n \geq 1$. Integrate by parts using $u = (\pm t + \frac{\pi}{2})$ and $v = \frac{\sin(nt)}{n}$.

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} r(t) \cos(nt) dt \\&= \frac{1}{\pi} \int_{-\pi}^0 (t + \frac{\pi}{2}) \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} (-t + \frac{\pi}{2}) \cos(nt) dt \\&= \frac{1}{\pi} \left((t + \frac{\pi}{2}) \frac{\sin(nt)}{n} \right) \Big|_{-\pi}^0 - \frac{1}{\pi} \int_{-\pi}^0 \frac{\sin(nt)}{n} dt \\&\quad + \frac{1}{\pi} \left((-t + \frac{\pi}{2}) \frac{\sin(nt)}{n} \right) \Big|_0^{\pi} + \frac{1}{\pi} \int_0^{\pi} \frac{\sin(nt)}{n} dt \\&= \frac{1}{\pi} (0) + \frac{1}{\pi} \frac{\cos(nt)}{n^2} \Big|_{-\pi}^0 + \frac{1}{\pi} (0) - \frac{1}{\pi} \frac{\cos(nt)}{n^2} \Big|_0^{\pi} \\&= \frac{1}{n^2 \pi} (1 - \cos(-n\pi)) + \frac{1}{n^2 \pi} (-\cos(n\pi) + 1) \\&= \frac{2}{\pi n^2} (1 - \cos(n\pi))\end{aligned}$$

Hence if n is even, $a_n = 0$ and if n is odd, $a_n = \frac{4}{\pi n^2}$.

Find Particular Solutions

Hence

$$y''(t) + 0.05y'(t) + 25y(t) = \sum_{n \geq 1, n \text{ odd}} \frac{4}{\pi n^2} \cos(nt)$$

equivalently,

$$y''(t) + 0.05y'(t) + 25y(t) = \sum_{n=0}^{\infty} \frac{4}{\pi(2n+1)^2} \cos((2n+1)t)$$

For each $n \geq 1$, n odd, find a particular solution y_n such that

$$y_n''(t) + 0.05y_n'(t) + 25y_n(t) = \frac{4}{\pi n^2} \cos(nt)$$

Use the method of undetermined coefficients and set

$$y_n = A_n \cos(nt) + B_n \sin(nt)$$

and substitute into the ODE to obtain

Find Particular Solutions and Combine

$$\begin{aligned}(25 - n^2)(A_n \cos(nt) + B_n \sin(nt)) + 0.05(-nA_n \sin(nt) + nB_n \cos(nt)) \\ = \frac{4}{\pi n^2} \cos(nt)\end{aligned}$$

Hence, comparing the coefficients of the sin and cos terms,

$$(25 - n^2)A_n + \frac{nB_n}{20} = \frac{4}{\pi n^2}$$

$$(25 - n^2)B_n = \frac{nA_n}{20}$$

$$A_n = \frac{20(25 - n^2)B_n}{n}$$

$$B_n = \frac{1}{5\pi n[(\frac{n}{20})^2 + (25 - n^2)^2]}$$

$$A_n = \frac{4(25 - n^2)}{\pi n^2[(\frac{n}{20})^2 + (25 - n^2)^2]}$$

To finish, $y_n = A_n \cos(nt) + B_n \sin(nt)$ and so

$$y(t) = \sum_{n \geq 1, n \text{ odd}} y_n = y_1 + y_3 + y_5 + \dots$$

Plot Series

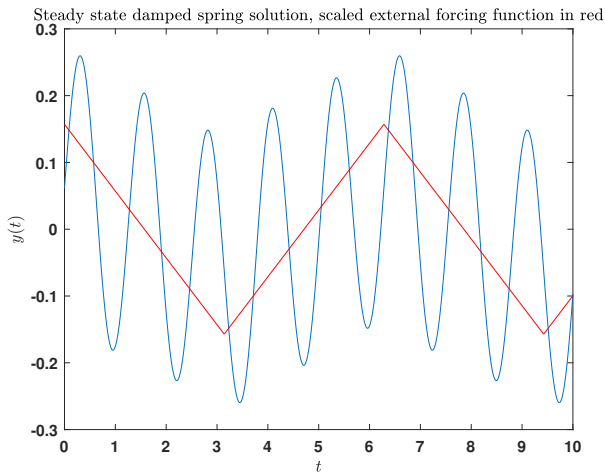


Figure: Steady state damped spring solution in blue, adding three terms up to $n = 5$; external triangular forcing function in red, scaled by 0.1.

Fourier Series

Example: Suppose we have a mass - spring - damper system with mass $m = 1$ kg, damping constant $c = 3$ kg/s, and spring constant $k = 7$ N/m. Suppose then the system has a periodic external force applied that is given by:

$$r(t) = t \quad \text{for } -1 < t \leq 1$$

$$\text{and } r(t+2) = r(t)$$

Find the steady state solution $y(t)$ (in metres).

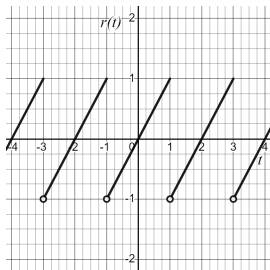


Figure: The periodic forcing function $r(t)$ (measured in newtons)

Fourier Series

Since mass $m = 1$, damping constant $c = 3$, and spring constant $k = 7$,

$$c^2 - 4mk = (3)^2 - 4(1)(7) < 0$$

and so we have damped oscillations that tend to zero as time $t \rightarrow \infty$.

Hence the homogeneous solution $y_h(t) \rightarrow 0$ as $t \rightarrow \infty$ and we need only find the particular solution.

Given

$$r(t) = t \quad \text{for } -1 < t \leq 1$$

$$\text{and } r(t+2) = r(t)$$

the Fourier series is

$$r(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\pi t) + b_n \sin(n\pi t))$$

where $a_n = 0$, $n \geq 0$, and $b_n = \frac{-2}{n\pi}$ if n is even and $b_n = \frac{2}{n\pi}$ if n is odd. See the last lecture for working. The Fourier series is

$$r(t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sin(n\pi t)}{n}$$

Find Particular Solutions

Hence

$$y''(t) + 3y'(t) + 7y(t) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2}{n\pi} \sin(nt)$$

For each $n \geq 1$, find a particular solution y_n such that

$$y_n''(t) + 3y_n'(t) + 7y_n(t) = (-1)^{n-1} \frac{2}{n\pi} \sin(nt)$$

Use the method of undetermined coefficients and set

$$y_n = A_n \cos(n\pi t) + B_n \sin(n\pi t)$$

and substitute into the ODE to obtain

Find Particular Solutions

$$\begin{aligned}& (7 - (n\pi)^2)(A_n \cos(n\pi t) + B_n \sin(n\pi t)) \\& + 3(-n\pi A_n \sin(n\pi t) + n\pi B_n \cos(n\pi t)) \\& = (-1)^{n-1} \frac{2}{n\pi} \sin(nt)\end{aligned}$$

Hence, comparing the coefficients of the sin and cos terms,

$$(7 - (n\pi)^2)A_n + 3n\pi B_n = 0$$

$$(7 - (n\pi)^2)B_n + -3n\pi A_n = (-1)^{n-1} \frac{2}{n\pi}$$

$$B_n = \frac{-(7 - (n\pi)^2)A_n}{3n\pi}$$

$$\frac{-(7 - (n\pi)^2)^2}{3n\pi} A_n - 3n\pi A_n = (-1)^{n-1} \frac{2}{n\pi}$$

$$A_n = (-1)^n \frac{2}{n\pi} \times \frac{3n\pi}{(7 - (n\pi)^2)^2 + (3n\pi)^2}$$

$$B_n = \frac{-(7 - (n\pi)^2)}{(7 - (n\pi)^2)^2 + (3n\pi)^2} \times (-1)^n \frac{2}{n\pi}$$

Find Particular Solutions and Combine

To finish,

$$y_n = A_n \cos(n\pi t) + B_n \sin(n\pi t)$$

and so the steady state solution is

$$y(t) = \sum_{n=1}^{\infty} A_n \cos(n\pi t) + B_n \sin(n\pi t)$$

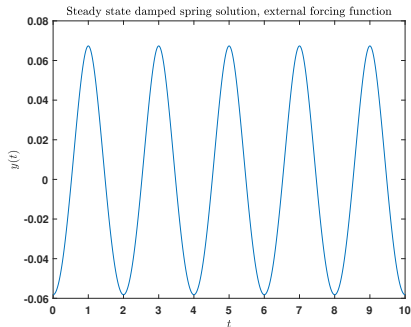


Figure: Steady state damped spring solution, external forcing function, 50 terms

Spring with External Periodic Forcing Function

Exercise: Find the steady state oscillations of

$$y''(t) + cy'(t) + y(t) = r(t)$$

assuming $0 < c < 2$ and

$$r(t) = \begin{cases} (t + \pi)t & \text{if } -\pi < t \leq 0 \\ -t(t - \pi) & \text{if } 0 < t \leq \pi \end{cases}$$

and $r(t + 2\pi) = r(t)$.

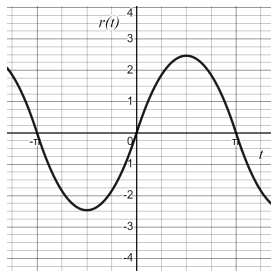


Figure: The periodic forcing function $r(t)$ (measured in newtons)

Text Book References

Chapters 11.1, 11.2

Spring with External Periodic Forcing Function: Exercise

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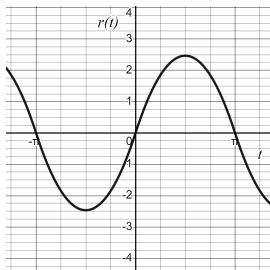


Figure: The periodic forcing function $r(t)$ (measured in newtons)

Working 1

Our function $r(t)$ has period 2π , so we can represent it as follows:

$$r(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nt) + b_n \sin(nt))$$

where:

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} r(t) dt$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} r(t) \cos(nt) dt$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} r(t) \sin(nt) dt$$

$$n \geq 1.$$

Working 2

The average height of the function is zero, by symmetry, so we expect $a_0 = 0$. Let us check.

$$\begin{aligned}a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} r(t) \, dt \\&= \frac{1}{2\pi} \int_{-\pi}^0 t^2 + \pi t \, dt - \frac{1}{2\pi} \int_0^{\pi} t^2 - \pi t \, dt \\&= \frac{1}{2\pi} \left(\frac{t^3}{3} + \frac{\pi t^2}{2} \right) \Big|_{-\pi}^0 - \frac{1}{2\pi} \left(\frac{t^3}{3} - \frac{\pi t^2}{2} \right) \Big|_0^{\pi} \\&= \frac{1}{2\pi} \left(0 + 0 + \frac{\pi^3}{3} - \frac{\pi^3}{2} \right) - \frac{1}{2\pi} \left(\frac{\pi^3}{3} - \frac{\pi^3}{2} + 0 - 0 \right) \\&= 0\end{aligned}$$

Since the function is odd, all the $a_n = 0$, $n \geq 1$.

Working 3

For $n \geq 1$,

$$b_n = \frac{1}{\pi} \int_{-\pi}^0 (t^2 + \pi t) \sin(nt) \, dt - \frac{1}{2\pi} \int_0^{\pi} (t^2 - \pi t) \sin(nt) \, dt$$

Setting $u = t$ and $v = -\frac{\cos(nt)}{n}$,

$$\begin{aligned} \int t \sin(nt) \, dt &= -t \frac{\cos(nt)}{n} + \int \frac{\cos(nt)}{n} \, dt \\ &= -t \frac{\cos(nt)}{n} + \frac{\sin(nt)}{n^2} + c \end{aligned}$$

Setting $u = t^2$ and $v = -\frac{\cos(nt)}{n}$,

$$\int t^2 \sin(nt) \, dt = -t^2 \frac{\cos(nt)}{n} + \int 2t \frac{\cos(nt)}{n} \, dt$$

Working 4

Setting $u = t$ and $v = \frac{\sin(nt)}{n}$,

$$\begin{aligned}\int t \cos(nt) \, dt &= t \frac{\sin(nt)}{n} - \int \frac{\sin(nt)}{n} \, dt \\ &= t \frac{\sin(nt)}{n} + \frac{\cos(nt)}{n^2} + c\end{aligned}$$

so

$$\begin{aligned}\int t^2 \sin(nt) \, dt &= -t^2 \frac{\cos(nt)}{n} + \int 2t \frac{\cos(nt)}{n} \, dt \\ &= -t^2 \frac{\cos(nt)}{n} + \frac{2}{n} \int t \cos(nt) \, dt \\ &= -t^2 \frac{\cos(nt)}{n} + \frac{2t \sin(nt)}{n^2} + \frac{2 \cos(nt)}{n^3} + c\end{aligned}$$

Working 5

So

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^0 t^2 \sin(nt) \, dt + \int_{-\pi}^0 t \sin(nt) \, dt \\ &\quad - \frac{1}{\pi} \int_0^{\pi} t^2 \sin(nt) \, dt + \int_0^{\pi} t \sin(nt) \, dt \\ &= \frac{2}{\pi} \int_{-\pi}^0 t^2 \sin(nt) \, dt + \int_{-\pi}^{\pi} t \sin(nt) \, dt \\ &= \frac{2}{\pi} \left(-t^2 \frac{\cos(nt)}{n} + \frac{2t \sin(nt)}{n^2} + \frac{2 \cos(nt)}{n^3} \right) \Big|_{-\pi}^0 \\ &\quad + \left(-t \frac{\cos(nt)}{n} + \frac{\sin(nt)}{n^2} \right) \Big|_{-\pi}^{\pi} \\ &= \frac{2}{\pi} \left(\frac{2}{n^3} + \frac{\pi^2 \cos(-n\pi)}{n} - \frac{2 \cos(-n\pi)}{n^3} \right) \\ &\quad + \left(\frac{-\pi \cos(n\pi)}{n} + \frac{-\pi \cos(-n\pi)}{n} \right) \end{aligned}$$

Working 6

If n is even

$$\begin{aligned}b_n &= \frac{2}{\pi} \frac{\pi^2}{n} - \frac{2\pi}{n} \\ &= 0\end{aligned}$$

If n is odd

$$\begin{aligned}b_n &= \frac{2}{\pi} \left(\frac{4}{n^3} - \frac{\pi^2}{n} \right) + \frac{2\pi}{n} \\ &= \frac{8}{\pi n^3}\end{aligned}$$

So

$$r(t) = \sum_{n=1, n \text{ odd}}^{\infty} \frac{8}{\pi n^3} \sin(nt) = \frac{8}{\pi} \sum_{n=1, n \text{ odd}}^{\infty} \frac{\sin(nt)}{n^3}$$

Working 7

For $n \geq 1$, n odd, solve

$$y_n'' + cy_n' + y_n = \frac{8}{\pi} \frac{\sin(nt)}{n^3}$$

Find a particular solution. Set

$$y_n = A_n \cos(nt) + B_n \sin(nt)$$

and find A_n and B_n . Differentiate to obtain

$$\begin{aligned} (1 - n^2)(A_n \cos(nt) + B_n \sin(nt)) + c(-nA_n \sin(nt) + nB_n \cos(nt)) \\ = \frac{8}{\pi} \frac{\sin(nt)}{n^3} \end{aligned}$$

Match the coefficients of the sine and cosine terms.

$$(1 - n^2)A_n + cnB_n = 0$$

so

$$B_n = \frac{(1 - n^2)A_n}{cn}$$

Working 8

and

$$(1 - n^2)B_n - cnA_n = \frac{8}{\pi n^3}$$

so

$$A_n \left(\frac{(1 - n^2)(1 - n^2)}{cn} \right) - A_n \frac{(cn)^2}{cn} = \frac{8}{\pi n^3}$$

so

$$A_n = \frac{8cn}{(\pi n^3) ((1 - n^2)^2 - (cn)^2)} = \frac{8c}{(\pi n^2) ((1 - n^2)^2 - (cn)^2)}$$

and

$$\begin{aligned} B_n &= \frac{8cn(1 - n^2)}{(cn)(\pi n^3) ((1 - n^2)^2 - (cn)^2)} \\ &= \frac{8(1 - n^2)}{(\pi n^3) ((1 - n^2)^2 - (cn)^2)} \end{aligned}$$

Working 9

Consider the damping for the unforced system. Since mass $m = 1$, damping constant $0 < c < 2$, and spring constant $k = 1$,

$$c^2 - 4mk < 0$$

and so we have damped oscillations that tend to zero as time $t \rightarrow \infty$. Hence the homogeneous solution $y_h(t) \rightarrow 0$ as $t \rightarrow \infty$ and we need only find the particular solution. To finish,

$$y_n = A_n \cos(nt) + B_n \sin(nt)$$

and so the steady state solution is

$$y(t) = \sum_{n=1, n \text{ odd}}^{\infty} A_n \cos(nt) + B_n \sin(nt)$$

Working 10

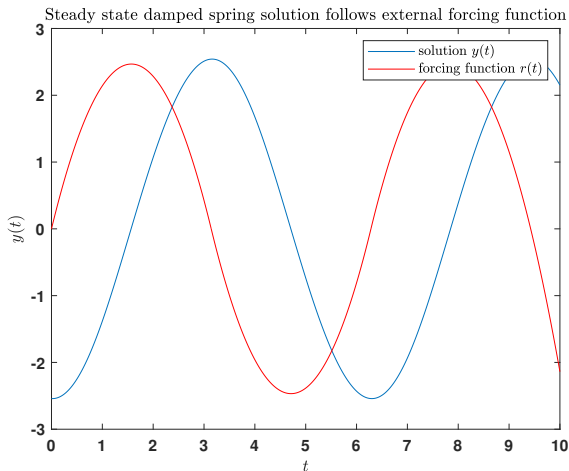


Figure: Steady state damped spring solution, external forcing function